

Migrate to the Edge

VMware, AWS, Azure, GCP → Edge KVM/KubeVirt with hyper2kvm + hypersdk

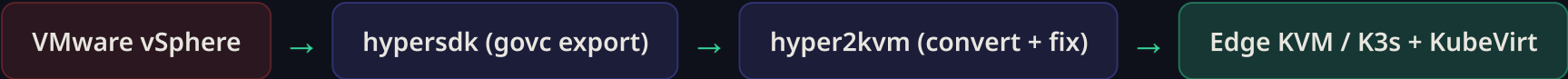
Move workloads from expensive data center hypervisors and cloud platforms to lightweight edge infrastructure. 10 source platforms, zero licensing cost, air-gap ready, single-server deployable. The complete edge migration toolkit.

hyper2kvm + hypersdk | Apache 2.0 Open Source

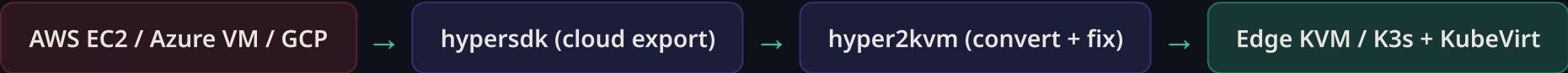
Every Source → Edge KVM

hypersdk exports from 10 platforms. hyper2kvm converts and deploys to edge infrastructure.

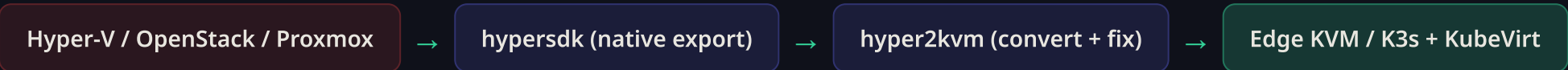
VMware → Edge



AWS / Azure / GCP → Edge



Hyper-V / OpenStack / Proxmox → Edge



All 10 Source Platforms

Source	Export Method	Edge Target	Use Case
VMware vSphere	OVA/OVF via govc, VDDK, HTTPS	KVM or K3s+KubeVirt	VMware exit at branch offices
AWS EC2	EBS snapshot, AMI export	On-prem edge KVM	Cloud repatriation to edge
Microsoft Azure	Managed disk snapshot export	Edge KVM or K3s	Reduce Azure egress costs
Google Cloud	Persistent disk, GCS staging	Edge KVM	Data sovereignty at edge
Hyper-V	VHD/VHDX direct copy	Linux KVM edge server	Windows-to-Linux edge migration
Oracle Cloud (OCI)	Boot volume export	Edge KVM	OCI exit, edge deployment
OpenStack	Glance image, Cinder volume	K3s + KubeVirt	Consolidate private cloud to edge
Alibaba Cloud	ECS disk export, OSS staging	Edge KVM	APAC edge deployments
Proxmox VE	API disk export	K3s + KubeVirt	Proxmox-to-K8s modernization

Why Move to the Edge?

Centralized data centers and cloud are expensive, slow, and fragile for distributed workloads.

VMware Licensing Crisis

Broadcom's acquisition of VMware caused 2-10x license increases. Running vSphere at 50 branch offices now costs **\$500K-\$2M/year** in software alone. Edge sites don't need enterprise hypervisor overhead.

Cloud Egress Costs

Edge workloads generate data locally but cloud VMs pay **\$0.09/GB egress**. A factory with 10TB/month of sensor data pays \$10,800/year just to move data out of AWS. On-prem edge: \$0.

Latency Requirements

Manufacturing SCADA needs **<10ms** response. Retail POS needs instant transaction processing. Healthcare imaging needs real-time access. Cloud round-trip: 20-100ms. Local KVM: **<1ms**.

Data Sovereignty

GDPR, HIPAA, and industry regulations require data to stay on-premises. Cloud storage may cross borders. Edge KVM keeps data **physically at the site** — compliant by design.

Network Unreliability

Remote sites (oil rigs, mines, ships, rural clinics) have intermittent connectivity. Cloud-dependent VMs go down when the link drops. Edge KVM **runs autonomously** — no internet required.

Cost Multiplier Effect

Every edge site multiplies licensing. 50 sites x vSphere + vCenter = financial disaster. **K3s + KVM = \$0 per site**. Scale to 500 sites with no software cost increase.

The answer: Move workloads from centralized VMware/cloud platforms to lightweight edge infrastructure powered by KVM + K3s + KubeVirt. hyper2kvm + hypersdk automates the entire journey.

The Edge Stack

What runs at each edge site — lightweight, free, production-ready.

Option A: Pure KVM

- Any Linux distro (Fedora, RHEL, Ubuntu)
- libvirt + KVM (built into kernel)
- hyper2kvm generates libvirt XML
- virsh define + start
- Simplest: 1 server, 0 dependencies**
- Best for: 1-5 VMs per site

Option B: K3s + KubeVirt

- K3s (single binary, 512MB RAM)
- KubeVirt for VM workloads
- GitOps fleet management
- Container + VM convergence
- Cloud-native: K8s APIs everywhere**
- Best for: 5-20 VMs + containers

Option C: Hybrid

- Some VMs on standalone KVM
- Others on K3s + KubeVirt
- hyper2kvm dual-target pipeline
- Same tool handles both
- Flexible: mix per workload needs**
- Best for: mixed workload sites

Per-Site Cost Comparison

Component	VMware Edge	Cloud (AWS/Azure)	hyper2kvm Edge
Hardware	\$3,000 (server)	N/A (cloud)	\$1,500 (mini PC/NUC)
Hypervisor	\$6,000-15,000/yr (vSphere)	Included	\$0 (KVM in kernel)
Management	\$4,000/yr (vCenter share)	\$200/mo (monitoring)	\$0 (GitOps / free)
Kubernetes	N/A	\$2,400/yr (EKS/AKS)	\$0 (K3s)
Compute (5 VMs)	Included in HW	\$3,600/mo	Included in HW
Egress (1TB/mo)	\$0 (local)	\$1,080/yr	\$0 (local)
Migration tool	N/A	N/A	\$0 (Apache 2.0)
Annual cost/site	\$10,000-19,000	\$46,680	\$500 (HW amortized)

50 Edge Sites — 3-Year Savings

vs VMware: Save \$1.4M - \$2.8M | **vs Cloud:** Save \$6.9M

How It Works

End-to-end: export from VMware/cloud at HQ, convert, deploy to every edge site.

Step 1: Export from Source (at HQ or cloud)

From VMware vSphere

```
# Using zkvm TUI:
# 1. Open vSphere tab
# 2. Connect to vCenter
# 3. Browse VMs, select batch
# 4. Ctrl+R to export + convert

# Or CLI:
sudo h2kvmctl --config edge.yaml
```

From AWS / Azure / GCP

```
# hypersdk exports from any cloud:
# AWS: EBS snapshot → download
# Azure: Managed disk → export
# GCP: Persistent disk → GCS
# OCI: Boot volume → export

# Unified API for all 10 providers
```

Step 2: Convert + Fix (hyper2kvm pipeline)



Fix Applied	What It Does	Why It's Critical for Edge
fstab UUID repair	Converts device names to UUID-based mounts	Edge hardware has different device naming — UUIDs are stable
initramfs rebuild	Injects virtio_blk, virtio_net, virtio_scsi	VM won't boot on KVM without VirtIO drivers
GRUB update	Fixes root= parameter for KVM disk path	Boot fails without correct root device
VMware tools removal	Removes open-vm-tools packages	Prevents errors and wasted resources at edge
Network cleanup	Removes VMware/Hyper-V NIC configs	Edge server has different NIC — clean slate needed
Windows VirtIO	Injects disk/net/balloon drivers + registry fix	Windows VMs BSOD without VirtIO on KVM
SELinux relabel	Triggers autorelabel on first boot	Prevents SELinux denials on new hardware

Step 3: Deploy to Edge

Deploy to KVM (libvirt)

hyper2kvm generates optimized libvirt XML, runs `virsh define`, and performs automated boot verification. Ship the QCOW2 image to the edge site, deploy with one command.

Deploy to K3s + KubeVirt

hyper2kvm generates KubeVirt VirtualMachine manifest + PVC. Upload disk image via CDI. Push manifest to Git — ArgoCD/Flux deploys to all edge K3s clusters automatically.

VMware Exit to Edge

The most common scenario: replace vSphere at every branch office with free KVM.

Before: VMware at Every Site

The Pain

- vSphere license per site: \$6,000-15,000/yr
- vCenter license: shared or per-site
- Broadcom renewal: 2-5x price increase
- Each site needs ESXi-compatible hardware
- vSphere expertise needed at every site
- License audits and compliance burden
- 50 sites = \$500K-\$1M/yr just for software

After: KVM/K3s at Every Site

- KVM: \$0 (built into Linux kernel)
- K3s: \$0 (CNCF open source)
- No license renewals, ever
- Runs on any x86/ARM hardware
- Basic Linux admin skills sufficient
- Zero audit burden — Apache 2.0
- 50 sites = \$0/yr in software costs

Migration Workflow: 50-Site VMware Exit

Phase	Action	What hyper2kvm + hypersdk Does	Result
1	Discovery	zkvm TUI connects to vCenter, discovers all VMs across 50 sites. Shows CPU, RAM, disk, power state per VM.	Complete inventory of 250+ VMs
2	Assessment	<code>h2kvmctl doctor</code> validates edge hardware readiness. Checks disk space, KVM support, network, libvirt.	Pre-flight green light per site
3	Batch Export	hypersdk batch-exports VMs from vSphere via govc. Streaming progress with ETA. Handles VMDK, OVA, OVF.	All disk images downloaded
4	Convert + Fix	VMCraft engine: VMDK → QCOW2, rebuild initramfs, fix fstab, remove VMware tools, inject VirtIO (Windows).	Boot-ready QCOW2 images
5	Deploy	Generate libvirt XML or KubeVirt manifests. Ship images to edge sites. <code>virsh define</code> OR <code>kubectl apply</code> .	VMs running on edge KVM/K3s
6	Verify	Automated boot test per VM. Webhook notification to Slack/Teams on success or failure.	Confirmed working at every site

Timeline: 5 sites/week = all 50 sites migrated in 10 weeks. One engineer with hyper2kvm can handle the entire project.

Cloud Repatriation to Edge

Bring workloads home from AWS, Azure, and GCP to on-premises edge infrastructure.

Why Repatriate to Edge?

Egress Costs Are Hidden Taxes

AWS charges \$0.09/GB outbound. A factory site pushing 5TB/month of data pays **\$5,400/year** just in egress — more than the edge hardware costs. On-prem: \$0 egress, forever.

Cloud Bills Only Go Up

Reserved instances expire. Spot instances get terminated. Pricing changes annually. Edge KVM has **fixed, predictable costs** — buy hardware once, run for 5+ years.

Latency-Sensitive Workloads

IoT gateways, real-time analytics, POS systems, SCADA controllers — these need **sub-10ms response**. Cloud adds 20-100ms of unavoidable latency. Edge KVM: <1ms.

Offline Capability

Edge sites must operate during network outages. Cloud VMs are useless without connectivity. Edge KVM **runs autonomously** — syncs when the link comes back.

Cloud-to-Edge Migration with hypersdk

Cloud Source	hypersdk Export Process	hyper2kvm Conversion	Edge Deployment
AWS EC2	EBS snapshot → S3 export → download VMDK/RAW	Convert to QCOW2, rebuild initramfs, fix fstab (Xen → VirtIO)	KVM at edge or K3s + KubeVirt
Azure VM	Managed disk snapshot → SAS URL → download VHD	VHD → QCOW2, remove Hyper-V agents, inject VirtIO	libvirt XML + virsh define
GCP	Persistent disk → GCS image export → download	RAW → QCOW2, fix boot for KVM, remove cloud-init	KubeVirt manifest + PVC upload
OCI	Boot volume → Object Storage → download	Convert + guest fixes for KVM compatibility	Automated deploy to edge K3s

Cost Savings: 20 VMs from Cloud to Edge (3 Years)

Cost Item	Cloud (AWS/Azure)	Edge KVM	Savings
Compute	\$720,000	\$60,000 (hardware)	\$660,000
Storage	\$180,000	\$15,000 (local NVMe)	\$165,000
Egress	\$90,000	\$0	\$90,000
Migration (one-time)	N/A	\$5,000 (hyper2kvm is free)	-\$5,000
3-Year Total	\$990,000	\$80,000	\$910,000

Industry Use Cases

Real-world edge migration scenarios powered by hyper2kvm + hypersdk.

Manufacturing — Factory Floor

- **Source:** VMware vSphere at 30 plants
- **Workloads:** SCADA, MES, quality control VMs
- **Edge target:** Ruggedized server + K3s + KubeVirt
- **Migration:** hypersdk exports from vCenter, hyper2kvm converts + deploys
- **Benefit:** \$280K/yr saved, sub-ms latency for PLC control
- **Advantage:** Air-gap ready — factories can't depend on internet

Retail — 200 Store Locations

- **Source:** AWS EC2 (POS backend) + VMware (inventory)
- **Workloads:** POS processing, inventory management, digital signage
- **Edge target:** Mini PC with K3s per store
- **Migration:** hypersdk pulls from AWS + vSphere, hyper2kvm converts centrally, GitOps deploys to all stores
- **Benefit:** \$1.2M/yr saved, transactions work during internet outages
- **Advantage:** One image converted, deployed to 200 sites via ArgoCD

Healthcare — 50 Clinics

- **Source:** Azure VMs (EMR system) + VMware (imaging)
- **Workloads:** Electronic medical records, PACS imaging, lab systems
- **Edge target:** HIPAA-compliant edge server + KVM
- **Migration:** hypersdk exports from Azure, hyper2kvm converts VHD → QCOW2, deploys to local KVM
- **Benefit:** \$480K/yr saved, data never leaves clinic (HIPAA)
- **Advantage:** Data sovereignty — patient records stay on-site

Energy — 100 Remote Substations

- **Source:** VMware at substations + GCP (analytics)
- **Workloads:** SCADA monitoring, grid analytics, telemetry
- **Edge target:** ARM-based gateway + K3s
- **Migration:** hypersdk exports from vSphere + GCP, hyper2kvm converts, deploys to ARM K3s clusters
- **Benefit:** \$800K/yr saved, works on satellite link or offline
- **Advantage:** Autonomous operation — no internet dependency

Telecom — Cell Tower Edge

- **Source:** VMware NFV + OpenStack VNFs
- **Workloads:** vRAN, packet core, edge CDN
- **Edge target:** Bare-metal K8s + KubeVirt at tower sites
- **Migration:** hypersdk exports from OpenStack, hyper2kvm converts VNFs, deploys as KubeVirt VMs
- **Benefit:** 60-80% NFV license savings, path to CNF

Government — Air-Gapped Sites

- **Source:** Hyper-V + VMware at classified facilities
- **Workloads:** Mission-critical apps, data processing
- **Edge target:** Hardened Linux + KVM (no internet ever)
- **Migration:** hyper2kvm runs fully offline — no internet, no phone-home, no telemetry
- **Benefit:** \$800K/yr saved, FedRAMP/STIG compliant

- **Advantage:** VNF → KubeVirt is step 1 toward cloud-native CNF

- **Advantage:** Single binary, USB-deployable, zero network requirement

Edge Fleet Management

Convert once at HQ, deploy to hundreds of edge sites with GitOps.

VMware / Cloud VMs



hyper2kvm (convert at HQ)



Container Registry



ArgoCD/Flux → 100 Edge Sites

Centralized Conversion

Run hyper2kvm at HQ: export from vSphere/cloud, convert VMDK → QCOW2, apply all guest fixes. **Convert once, deploy everywhere.** No need to run conversion tools at each edge site.

GitOps Distribution

Push KubeVirt manifests to Git. ArgoCD syncs to every K3s edge cluster. **One git push = deploy to 100 sites.** Rollback by reverting a commit. Full audit trail.

Zero-Touch Edge Setup

New site: plug in hardware, PXE boot, K3s auto-joins fleet. ArgoCD deploys VMs automatically. **No SSH needed. No on-site IT.** Ship the box, plug it in, done.

Central Monitoring

Prometheus + Grafana federated across all sites. **Single dashboard** for CPU, memory, disk, VM health across entire fleet. Alert on anomalies before they become outages.

Example: Edge Migration YAML Config

```
# edge-factory-vm.yaml - Convert VMware VM for edge deployment
cmd: vsphere
vcenter: vcenter.hq.corp
vs_action: export_vm
vs_vm: factory-scada-01
output_dir: ./edge-images
out_format: qcow2
flatten: true
compress: true           # Smaller image for shipping to edge
regen_initramfs: true    # Inject VirtIO for KVM
emit_domain_xml: true    # Generate libvirt XML (for pure KVM sites)
deploy_k8s: true         # Also generate KubeVirt manifest (for K3s sites)
boot_test: true          # Verify VM boots before shipping
```

Dual target in one run: `emit_domain_xml + deploy_k8s` generates both libvirt XML (for pure KVM edge sites) and KubeVirt manifests (for K3s edge sites) from a single conversion. Deploy the right format per site.

Why hyper2kvm for Edge

Technical advantages that make hyper2kvm the ideal edge migration tool.

Lightweight Engine

VMCraft = pure Python + qemu-nbd. No libguestfs appliance (500MB+). No supermin boot. Runs on minimal Linux installs, containers, even ARM. **~5MB footprint.**

Air-Gap Ready

Single binary + Python package. No internet needed — ever. No telemetry, no phone-home, no license server. **USB-deployable** to classified and disconnected sites.

10 Source Platforms

Export from VMware, AWS, Azure, GCP, Hyper-V, OCI, OpenStack, Alibaba, Proxmox, KubeVirt. **One tool for every source.** One learning curve for every migration.

Dual Target

Deploy to standalone KVM/libvirt **and/or** K3s + KubeVirt from the same pipeline. Mix targets per site. No other tool does both.

Boot Verification

Automated boot test after conversion. **Know it works before shipping to a remote site.** Don't discover boot failures at a location 500 miles away.

Interactive TUI

zkvm guides operators step-by-step. File browser, source auto-detection, live progress. **Remote site staff don't need CLI expertise.**

Job Scheduling

hypersdk schedules migrations for off-peak hours. **Carbon-aware scheduling** runs during low grid intensity. Cron, priority queues, retry policies.

Pre-Flight Doctor

`h2kvmctl doctor` checks 9 areas: disk space, KVM support, libvirt, network, permissions. **Validate edge hardware before migration day.**

\$0 Cost

Apache 2.0 license. No subscription, no per-VM fee, no per-site fee. **Scale from 1 to 1,000 edge sites with zero software cost increase.**

The Edge Migration Platform

Export from any source. Convert with full guest fixes. Deploy to any edge target.

Air-gap ready. Lightweight. Free forever.

hyper2kvm + hypersdk — the only tool built for edge-scale migration.

